

Phase 6: Project Testing

6.1 Testing Objectives

Verify that LegalEase works end-to-end: form input → FastAPI → Gemini → preview/edit → TXT/DOCX/PDF export, under valid, invalid and edge-case inputs.

6.2 Test Environment

Item	Details
OS	Windows 10/11 (or your OS)
Python	3.10+
Browser	Google Chrome
Backend	<code>uvicorn legalEaseAPI.main:app --reload</code> on port 8000
Frontend	<code>streamlit run frontend/app.py</code> on port 8501
AI model	Gemini (configured in <code>config.py</code>)

6.3 Test Types

- **Unit testing:** `sanitize_text`, `format_html_preview`, `format_docx`, `format_pdf`
- **API testing:** GET `/` and POST `/generate` via Swagger UI (`/docs`) or Postman

- **Integration testing:** Streamlit → FastAPI → Gemini
- **Functional / UI testing:** full user flow in the browser
- **Input-combination testing:** different document types, parties, terms and date formats

6.4 Test Cases

Fill the **Actual Result** and **Status** columns after running each test (attach screenshots in `docs/screenshots/`).

ID	Feature	Test Steps	Expected Result
TC-01	Form validation	Click Generate with all fields empty	Warning: fill all fields; no AP
TC-02	Form validation	Fill 3 of 4 fields, click Generate	Warning shown; no AP
TC-03	Generation	Freelance Work Contract, Jane Doe / TechNova Inc., 4 terms, April 15, 2025	Success message; document parties, terms, date, se
TC-04	Preview	After TC-03	Styled, scrollable preview document
TC-05	Edit	Click Edit, change text (e.g. add a line), export	Edit reflected in preview exports
TC-06	TXT export	Click Download as .TXT	File <code>freelance_work_co</code> opens in Notepad with
TC-07	DOCX export	Click Download as .DOCX	Opens in Word; logo, T Roman, terms table, fo
TC-08	PDF export	Click Download as .PDF	Opens in viewer; logo h footer on every page; n text wraps correctly
TC-09	Sanitization	Enter terms with curly quotes and long dashes	No crash; PDF and DO generated cleanly

TC-10	Document types	Try NDA, Lease Agreement, Employment Offer Letter	Each generates a relevant structured document
TC-11	Health check	Open <code>http://127.0.0.1:8000/</code>	Welcome JSON message
TC-12	Backend down	Stop Uvicorn, click Generate	Friendly error message; not crash
TC-13	Invalid API key	Use a wrong <code>GEMINI_API_KEY</code>	API returns 500 with relevant message; UI shows error
TC-14	Date formats	Try <code>April 10, 2025</code> and <code>10/04/2025</code>	Both accepted and reflected in document
TC-15	Long input	Paste 15+ terms	Document generated; all terms appear; PDF paginates
TC-16	File naming	Type "Lease Agreement"	Files named <code>lease_agreement.pdf</code>

6.5 API Test Examples

Health check

```
curl http://127.0.0.1:8000/
```

Generate

```
curl -X POST http://127.0.0.1:8000/generate \
  -H "Content-Type: application/json" \
  -d '{"document_type": "NDA", "parties": "Alice (Freelancer"
```

Missing field (expect 422)

```
curl -X POST http://127.0.0.1:8000/generate -H "Content-Type: application/json"
```

6.6 Observations from Initial Test Run

(Freelance Work Contract)

The sample run (Jane Doe / TechNova Inc., April 15, 2025) confirmed:

- Generation, preview, edit, and TXT/DOCX/PDF downloads all work
- Edits made in the text area (test line added) appeared in the TXT, DOCX and PDF outputs
- DOCX contains the logo and footer; PDF shows logo and footer on all 4 pages

Issues observed and recommended fixes:

#	Issue	Cause	Fix
1	Markdown heading <code>## Freelance Work Contract</code> appears literally in preview, DOCX and PDF	Gemini returns markdown	Prompt asks for plain text; <code>_clean_line</code> strips <code>#</code> ; <code>sanitize_text</code> strips <code>**</code>
2	Bracketed placeholders remain (e.g. [Jane Doe's Address], [State], [Dollar Amount])	User did not supply these details	Expected behaviour; add optional address/ jurisdiction/amount inputs, and tell users to complete placeholders before use
3	Wide lines in the TXT file need horizontal scrolling in Notepad	Long unwrapped paragraphs	Enable word wrap in Notepad or insert line wraps (<code>textwrap</code>) when exporting TXT
4	Non-Latin characters (e.g. ₹) replaced by ? in PDF	Classic FPDF is Latin-1 only	Use <code>fpdf2</code> with a Unicode TTF font (e.g. DejaVu)

5	Model name may become unavailable	Google retires older models	Change <code>GEMINI_MODEL</code> in <code>.env / config.py</code>
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6.7 Bug Tracker

Bug ID	Description	Severity	Status
B-01	Markdown symbols in output	Medium	Fixed (prompt + cleanup)
B-02	PDF encoding error on curly quotes	High	Fixed (<code>sanitize_text</code> , <code>_latin1</code>)
B-03	Unicode characters lost in PDF	Low	Open / future improvement
B-04	Unhandled exception when backend is offline	Medium	Fixed (<code>try/except</code> with <code>st.error</code>)

6.8 Test Summary

Metric	Value
Total test cases	16
Passed	
Failed	
Blocked	

Conclusion: